

Research report after round 1. The run is still in progress: this is the report text as it stands after this round, text only, with no figures. The results and the writing are not final.

Framing

Current safety evaluation requires running full benchmarks with hundreds of prompts, taking minutes per model, or comparing two models (base versus aligned). This makes it impractical to screen random models from HuggingFace for safety. We investigated whether internal activation properties -- measurable from just a few prompts without a reference model -- can separate safe from unsafe models. We call a metric that requires only one model, no reference, a *single-model metric*. The question is whether safety fine-tuning creates detectable changes in how models process harmful versus benign content at the level of hidden states, and whether those changes are large enough to serve as a fast screening metric.

Iteration 1: Metric Exploration on Qwen3-4B Lineage

Strategy

We started with one model lineage where the safety gradient is clearest: Qwen3-4B-Base (no safety tuning), Qwen3-4B-SafeRL (official safety RL model), and Huihui-Qwen3-4B-abliterated-v2 (a community uncensored model, which we call the *abliterated model*). The instruct, SafeRL, and abliterated models share a chat template, making them directly comparable; the base model uses a different format and is kept separate. We designed ten candidate metrics -- nine reading hidden states or weights, one *logit-only baseline* (reading only final logits, no internal activations) -- and tested them on 10 harmful and 10 benign prompts drawn from JBB-Behaviors and XSTest datasets. The metrics span different hypotheses about what safety tuning changes: activation geometry (norms, entropy, sparsity), *cross-layer structure* (information flow and direction consistency between consecutive layers), weight properties (spectral norms), and output confidence (logit gaps).

Artifact 1: Safety Prompt Dataset Curation

We curated a safety prompt dataset from three benchmark sources, selected from six candidates based on category diversity, example count, balanced harmful/benign split, and confirmed provenance (published papers, over 100 downloads on HuggingFace).

Table 1: Dataset summary

Dataset	Examples	Categories	Harmful	Benign	Source
XSTest	450	18	225	225	Paul/XSTest
PKU-SafeRLHF	73,907	20	~37,000	~37,000	PKU-SafeRLHF
JBB-Behaviors	100	10	50	50	JailbreakBench/JBB-Behaviors
Total	74,457	20	~37,275	~37,182	

All prompts are English, under 512 tokens, with binary safety labels. Each example includes metadata for category, source dataset, fold assignment, and task type. For the experiment, we sampled 10 harmful and 10 benign prompts from JBB-Behaviors and XSTest.

Artifact 2: Safety Metric Evaluation

Implementation

The experiment loads one model at a time in bf16 on CPU, captures residual stream activations at 8 layers spaced evenly across the depth (0, 4, 8, 12, 16, 20, 24, 28) using forward hooks, and computes all metrics for each individual prompt. Each model takes approximately 40 seconds to process 20 prompts. *Random-weight controls* verify that signals are not spurious by replacing activations with random noise and recomputing each metric.

Metric definitions

1. **ATC (Activation Trajectory Consistency):** Cosine similarity between harmful and benign activations across layers. Higher means more similar trajectories.
2. **WSSS (Weight Spectral Signature):** Average spectral norm of MLP up_proj weights across first 4 and last 4 layers. Single scalar per model.
3. **CLIF (Cross-Layer Information Flow):** Stability of activation norm ratios across consecutive layers, measured as $1/(1 + CV)$ where CV is coefficient of variation.
4. **MAS (MLP Activation Sparsity):** Fraction of activation values below 1% of maximum absolute value.
5. **Logit (Logit Gap Baseline):** Top-2 logit margin at last token position.
6. **ANG (Activation Norm Gap):** Mean Euclidean norm of activations across layers.
7. **AEN (Activation Entropy):** Shannon entropy of normalized absolute activation distribution.
8. **LWCD (Layer-by-Layer Cosine Distance):** Average cosine distance between mean harmful and mean benign activations, computed separately for each layer.
9. **MLPS (MLP Output Sparsity):** Fraction of activation values within 0.1 standard deviations of zero.
10. **RSD (Residual Stream Direction):** Average cosine similarity between consecutive layer activations.

Results: Metric Ranking

Table 2: Metric ranking by average absolute Cohen's d

Rank	Metric	avg	d	d(SafeRL hvsb)	d(Base hvsb)	d(Abliterated hvsb)	Separation Score
1	AEN	6.5335	5.1503	9.1607	5.2895	-2.0748	
2	ANG	3.9330	-2.9932	-5.5532	-3.2526	-1.4097	
3	CLIF	1.5545	1.2394	2.0735	1.3505	-0.4726	
4	MLPS	1.4195	-1.3620	-2.4128	-0.4837	-0.0863	
5	MAS	1.2437	1.3569	1.1013	1.2728	0.1698	
6	RSD	0.6390	-0.5887	-0.6779	-0.6505	-0.0755	
7	Logit	0.2023	-0.1326	-0.0990	0.3752	-0.1045	
8	ATC	3.2e-14	-9.5e-14	0.0	0.0	9.5e-14	
9	LWCD	0.0	0.0	0.0	0.0	0.0	
10	WSSS	0.0	0.0	0.0	0.0	0.0	

Cohen's d measures the standardized difference between harmful and benign prompt scores within each model. A large $|d|$ means the metric distinguishes harmful from benign prompts well. The separation score measures how much better SafeRL separates harmful versus benign compared to

the other models.

[Figure fig1: generated in the report phase, not in this draft]

Results: Mean Values per Model

Table 3: Mean metric values per model and prompt type

Metric	SafeRL (harmful)	SafeRL (benign)	Base (harmful)	Base (benign)	Abliterated (harmful)	Abliterated (benign)
ATC	0.9950	0.9950	0.9915	0.9915	0.9951	0.9951
WSSS	2.8211	2.8211	2.7626	2.7626	2.7367	2.7367
CLIF	0.3791	0.3645	0.4092	0.3881	0.3792	0.3644
MAS	0.9999	0.9999	0.9999	0.9998	0.9999	0.9999
Logit	1.1563	1.3125	0.5500	0.6000	1.4188	1.0375
ANG	301.26	406.71	247.54	330.39	299.76	404.51
AEN	11.3793	10.7542	11.5705	10.9942	11.3594	10.8248
LWCD	0.0044	0.0044	0.0029	0.0029	0.0024	0.0024
MLPS	0.9474	0.9605	0.9196	0.9396	0.9334	0.9376
RSD	0.8358	0.8370	0.8348	0.8362	0.8356	0.8370

The key pattern: AEN and ANG show the largest gaps between harmful and benign prompts. For AEN, harmful prompts have higher entropy (more spread-out activations) than benign ones across all three models. For ANG, harmful prompts have lower activation norms than benign ones. The gap is largest for the Base model ($d = 9.16$ for AEN, $d = 5.55$ for ANG) and smallest for SafeRL ($d = 5.15$ for AEN, $d = 2.99$ for ANG), suggesting safety tuning reduces the activation difference between harmful and benign content.

[Figure fig2: generated in the report phase, not in this draft]

Results: Random-Weight Controls

Table 4: Random-weight control values

Metric	Base (real)	Base (random)	SafeRL (real)	SafeRL (random)	Abliterated (real)	Abliterated (random)
ATC	0.9915	-0.0001	0.9950	0.0003	0.9951	-0.0006
CLIF	0.3987	0.3895	0.3718	0.3627	0.3718	0.3823
MAS	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999
Logit	0.5750	0.5750	1.2344	1.2344	1.2281	1.2281
ANG	288.96	226.75	353.98	291.15	352.13	275.56
AEN	11.2823	11.2936	11.0667	11.0803	11.0921	11.0997
LWCD	0.0029	0.9971	0.0044	0.9889	0.0024	1.0010
MLPS	0.9296	0.9249	0.9540	0.9488	0.9355	0.9307
RSD	0.8355	-0.0006	0.8364	-0.0002	0.8363	-0.0007

The random controls show that ATC, LWCD, and RSD collapse to values indistinguishable from zero or random noise when activations are randomized, confirming they read actual activation structure. ANG and AEN show different values for random versus real activations, confirming their signals come from the data. The logit baseline is unchanged because randomizing hidden states does not

change the logit gap computation (which reads only the final logits).

Placebo Test

An independent audit script (audit.py) recomputed all Cohen's d values using scipy.stats instead of the manual formula in method.py, confirming all stored numbers match exactly. It also ran a placebo test by shuffling harmful/benign labels and recomputing d. The shuffled labels produced lower $|d|$ for the strong metrics (AEN, ANG, CLIF, MAS, MLPS), confirming the signals are real and not artifacts of the label assignment.

Dead Ends

ATC (Activation Trajectory Consistency): Cohen's d values indistinguishable from zero across all models ($\text{avg}|d| = 3.2\text{e-}14$). The cosine similarity between harmful and benign activations is nearly identical for both prompt types. This killed the original hypothesis that safety tuning creates more consistent activation trajectories on harmful prompts. The metric reads the same value for harmful and benign prompts, so it cannot separate them.

LWCD (Layer-by-Layer Cosine Distance): Zero Cohen's d across all models. The mean harmful and mean benign activations at each layer are nearly identical in direction, producing the same cosine distance regardless of prompt type. Average distance computed layer by layer is not discriminative.

WSSS (Weight Spectral Signature): Zero Cohen's d because this metric produces a single scalar per model (average spectral norm of weight matrices), not a score computed for each individual prompt. It cannot compare harmful versus benign prompts within a model. The values differ slightly across models (SafeRL = 2.82, Base = 2.76, Abliterated = 2.74) but this is a property of the weights, not of how the model processes specific prompts.

RSD (Residual Stream Direction): Small negative separation score (-0.075). The direction consistency of residual stream activations across layers does not differ meaningfully between harmful and benign prompts. The Cohen's d values are moderate (around 0.6) but the separation score is negative, meaning SafeRL does not separate harmful from benign better than the other models.

Logit gap: Weak signal ($\text{avg}|d| = 0.20$). The top-2 logit margin shows only a small difference between harmful and benign prompts, and the separation score is negative. This confirms that logit-only metrics are insufficient for safety screening and that looking inside the model buys a substantial advantage.

What we have learned so far

Activation entropy (AEN) and activation norm gap (ANG) are the strongest single-model safety signals found so far, with effect sizes an order of magnitude larger than the logit-only baseline. AEN measures how spread out the activation distribution is -- harmful prompts produce more entropic (more uniform) activations than benign ones. ANG measures the magnitude of activations -- harmful prompts produce smaller norms. Both metrics work on all three models in the lineage, though the effect size is largest for the Base model and smallest for SafeRL, suggesting safety tuning reduces the activation difference between harmful and benign content.

The original hypothesis about activation trajectory consistency (ATC) was disconfirmed: cosine similarity between harmful and benign trajectories does not differ between prompt types. The

weight spectral signature (WSSS) is not usable as a metric that produces a score for each individual prompt. The metrics that work read the geometry of individual activations (entropy, norm, sparsity), not their relationships across prompts or layers.

The experiment ran on CPU in approximately 10 minutes total for all three models, demonstrating these metrics can provide fast safety estimates without GPU or full benchmark runs. The next step is to test these metrics on more model families and correlate them with independent safety benchmarks.

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